Lines in the Blank Sky: a Search of the SDSS Sky Fibres for Laser Emission, and the Galaxies Behind Them

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Abstract

Every BOSS and eBOSS plate of the Sloan Digital Sky Survey places some eighty fibres on positions with no catalogued source, to measure the sky. A point source of line emission too faint in broad bands for the imaging that chose those positions, a transmitter on its own or beside a star too faint to catalogue, would appear in one of them as an unresolved emission line that the plate's other sky fibres lack. We search the 381,943 sky-fibre spectra of the 3853 plates of DR17, from 3600 to 10400 Å at a resolution near 2000, for such lines: matched-filter peaks at 6σ, off the airglow lines, as narrow as the instrument and absent from the plate's other sky fibres. Absorption-like dips of the same shape, which no transmitter makes, are put through every test the peaks are and serve as the null. A line that one fibre holds at one wavelength at other positions on other plates is the detector's: 762 of 3217 clean peaks and 1813 of 4108 dips. Two or more peaks at one redshift, at the flux ratios a galaxy's spectrum fixes, mark 156 fibres, where the same test finds 4 among the dips and 12 ± 4 among shuffled peaks: 154 galaxies at $z = 0.02$ to 1.07, six of them seen twice on different plates, 122 with a Legacy Surveys DR10 source within $1.5''$ of median $r = 23.1$, and two fibres of Galactic nebular lines. At 8σ, with the spectra that hold clean lines of both signs set aside, 485 peaks and 336 dips remain. Legacy Surveys sources lie within $1.5''$ of 38 per cent of the peaks against 4.4 per cent at positions $30''$ north, and of the dips less often than at those positions, so about 160 of the peaks are single lines of galaxies, and the other 323 fall short of the 336 dips. Read in each exposure, 359 peaks and 241 dips hold through the frames, and net of their counterparts the 202 persistent peaks fall short of the 253 persistent dips. The line that recurs most often at one sky position, at 4732.6 Å on 54 plate-nights over seven years, is resolved and alone in the stack of the 94 spectra there: Lyα of a galaxy at $z = 2.893$. The median line flux at 8σ, 8×10^{-17} erg cm^{-2} s^{-1}, is that of a transmitter aimed at Earth through a 10 m aperture at 37 W from 10 pc and 3.7 kW from 100 pc, and with 0.68 of such lines recovered, the 0.093 deg^2 the fibres cover bound such transmitters to fewer than about 850 per square degree.

1 Introduction

Spectroscopic searches for laser emission look for an unresolved line in the spectrum of a star [1, 2], on the reasoning that a transmitter lives near one. A transmitter need not: one in free space, or beside a star too faint for any catalogue, has no spectrum of its own taken unless a fibre happens to fall on it. Large fibre surveys do place fibres on empty sky, to measure the sky they subtract from every other spectrum, and the Sloan Digital Sky Survey's BOSS [7] and eBOSS [8] spectrographs [9] recorded some eighty such spectra on each of thousands of plates. Their positions were chosen to hold nothing in the survey's imaging, so a source there is fainter in broad bands than that imaging reaches, and a line from it stands alone.

The same spectra catch what the imaging missed. A galaxy whose light is mostly in a few emission lines is faint in broad bands and bright in its lines, and a sky fibre that lands on one records them at one redshift; such lines are the commonest astrophysical source a laser search

of blank sky must recognise. This paper searches every sky fibre of DR17's BOSS and eBOSS plates for unresolved emission lines, separates the galaxies by their redshifts and by the sources the imaging holds at the fibres, and measures the rest, in the combined spectra and in each exposure, against the same search for dips, put through the same tests.

1.1 Contributions

1. A search of 381,943 sky-fibre spectra on 3853 plates for unresolved emission lines absent from the plate's other sky fibres, with absorption-like dips put through every test as the null.
2. A catalogue of the 154 galaxies whose lines the search finds at one redshift, with their Legacy Surveys counterparts.
3. A bound on transmitters in blank sky from the excess of lines over the null, drawn again on the lines that hold in every exposure, and the per-line data, peaks and dips, from which every number here follows.
4. The lines that recur at one sky position over several plate-nights, among them a Lyα emitter at $z = 2.893$ found by its recurrence alone.

2 Data and method

Data. DR17 [10] holds, for each BOSS and eBOSS plate and night, an spPlate file with the flux, inverse variance and sky-model spectrum of every fibre on a common grid of 10^{-4} in $\log_{10}\lambda$, and the plug map that names each fibre's target type. The fibres of type SKY give the sky spectra, whose rows we read from the archive by byte range, one plate at a time: 4131 files of the 3853 plates.

Detection. Each sky spectrum is scaled to its own robust noise: the flux times the square root of the inverse variance, divided by 1.4826 times its median absolute deviation, with pixels the pipeline masks given zero weight. A filter matched to a Gaussian of $\sigma = 1.1$ pixels, the instrumental profile, gives the significance of a line at each pixel, and a peak at 6σ or more is kept unless one of four tests sets it aside: a masked pixel within one of it; an airglow line, where the plate's median sky model exceeds three times its running continuum, widened by two pixels; two or more other sky fibres of the same plate at 3σ within a pixel, which marks a residual of the sky subtraction; or a width, from the peak's second moment, outside 0.6 to 2 times the profile. A peak that passes all four is clean. The same search run on the negated spectra gives dips of the same shape. Each peak and dip that passes the first, second and fourth tests is kept with the number of other fibres at 3σ beside it, so the rule of one other fibre, stricter and less complete, is applied to the same detections for comparison.

Tests after detection. Three tests act on the clean lines of each sign alike. A line that the same fibre holds within three pixels on another plate, at a sky position more than $1.5''$ away, belongs to the detector; plates of one field can return a fibre to the same position, and a line it holds there is the sky's. Two or more lines of one fibre within $4 \times 10^{-4}(1 + z)$ of the vacuum wavelengths of [O II] 3727, Hβ, [O III] 4959 and 5007, Hα, [N II] 6583, [S II] 6716 and 6731 or Lyα at one redshift above -0.002, which admits Galactic nebulae, mark a galaxy, provided that one of them is [O II], Hβ, [O III] 5007 or Lyα, or Hα beside one of those, and that [O III] 4959 appears only with 5007 wherever 5007 falls in the spectrum, at a flux ratio 5007/4959 of 1.5 to 5 about the 2.98 of the transition probabilities. The dips, which no galaxy makes, measure how often residuals meet this test by chance. A spectrum that holds clean lines of both signs outside the first test is defective,

and its lines of either sign are set aside. Each galaxy's position is matched within 1.5″ to the Legacy Surveys DR10 catalogue [11] and to DESI DR1's redshifts [12] through the Astro Data Lab [13].

Counterparts. A galaxy too faint in its other lines shows one line, and the imaging may hold it. For every line at 8σ that the tests leave, peaks and dips alike, the Legacy Surveys DR10 sources within 1.5″ of the fibre and of the position 30″ north of it are read from the Astro Data Lab and the Legacy Survey viewer's catalogue service; the excess at the fibres over the northern positions counts the lines that come from catalogued sources.

Exposures. An spPlate spectrum combines several exposures, each recorded in a calibrated frame of the blue camera and one of the red. Every line at 8σ that the tests leave, and every galaxy line at 8σ, is measured again in each frame that covers it, with a Gaussian of the frames' instrumental profile, $\sigma = 1.2$ pixels, placed within three pixels of the line's wavelength. A line that one frame carries, at 5σ or more with over 80 per cent of the summed squared significance, is a cosmic ray or a detector event; one at 4σ in one camera and below 2σ in the other, where the cameras overlap, is that camera's; one at 2.5σ or more in two or more frames is persistent; the rest are unclear, and a line that fewer than two frames cover is left unjudged.

Recurring positions. A plate observed on several nights keeps its sky fibres where they are, and later plates of a field return to some of its positions. Clean lines at one position, friends of friends within 1.5″, recur when they fall on two or more plate-nights within 3×10^{-4} in $\log \lambda$; a line that recurs at the same wavelength at another position as well is the sky's. The dips, counted the same way, are the null.

3 Results

3.1 Peaks and dips

The 381,943 sky spectra hold 115,169 emission-like peaks at 6σ and 75,591 absorption-like dips. The four tests leave 3217 peaks and 4108 dips, of which 970 and 1479 reach 8σ. Table 1 follows both signs through the tests after detection, under the two-fibre rule and the one-fibre rule.

3.2 Lines fixed to a fibre

A line that one fibre holds on several plates at one wavelength and at different positions belongs to the detector, not the sky: 762 clean peaks, 287 at 8σ, in 131 fibres, and 1813 dips, 1114 at 8σ, recur in the same fibre within three pixels on other plates. Fibre 832 near 9280 Å carries 42 of the peaks, on the red camera where the airglow forest is densest, and fibres 10 at 6600 Å and 210 at 4460 Å carry 19 and 17. The test within a plate cannot find these, since each plate holds the fibre once, and three quarters of the dips at 8σ are of this kind. The condition on position keeps on the sky 36 peaks and one dip that a fibre holds where a later plate of the field returns it to the same position.

3.3 Galaxies

The redshift test marks 156 fibres, against 11.6 ± 3.5 when the peaks' wavelengths are shuffled among the 325 fibres that hold two or more, each fibre keeping its count, over 200 realisations, and 4 among the dips. Without its requirements on Hα and the [O III] ratio it would mark 187, and 30 among the dips: Hα with [N II] or [S II] alone (16 fibres among the peaks and 11 among the dips), [O III] 4959 without 5007 (11 and 6) and the [O III] doublet at a ratio outside 1.5 to 5 (4 and 9), the forms the residuals of the sky subtraction take. Some of the fibres set aside repeat features at

Table 1: Peaks and dips at each stage, under the rule of two other fibres and of one. Each later row applies its test to both signs; the counterpart rows give the lines with a Legacy Surveys DR10 source within 1.5″ at the fibre and at the position 30″ north, among those the catalogue covers, and the persistent rows the lines of the two-fibre rule that hold in two or more of the exposures' frames.

	Two other fibres		One other fibre	
	Peaks	Dips	Peaks	Dips
Clean at 6σ	3217	4108	1633	1874
fixed to a fibre	762	1813	242	847
at a galaxy's redshift	369	22	144	8
with lines of both signs	30	38	17	14
the rest	2056	2235	1230	1005
Clean at 8σ	970	1479	521	783
fixed to a fibre	287	1114	119	537
at a galaxy's redshift	188	19	78	6
with lines of both signs	10	10	9	4
the rest	485	336	315	236
source at the fibre	181	10	138	6
source 30″ north	21	28	12	15
net of counterparts	323	354	188	245
persistent	359	241		
net of counterparts	202	253		

the same observed wavelengths on other plates, at 9842 and 10097 Å, 9842 and 9886 Å and 9906 and 10113 Å among others, the residuals the subtraction leaves between the strong OH lines of the red camera.

Two fibres hold lines at zero redshift, the nebular emission of the Galaxy where a fibre crosses it. The other 154 are galaxies (Table 2) at $z = 0.016$ to 1.069, median 0.441 and quartiles 0.33 and 0.64, found by [O III] 5007 in 139, [O II] in 71, Hα in 56, [O III] 4959 in 35, Hβ in 32 and [S II] and [N II] in one each. The 19 found by the [O III] doublet alone have flux ratios 5007/4959 of 1.8 to 4.1, median 2.7, about the 2.98 of the transition probabilities. The twelve whose lines all lie above 8300 Å, in the red forest, are each Hβ with [O III] or the [O III] doublet. The [O III] 5007 fluxes run from 3.0×10^{-17} to 9.8×10^{-16} erg cm^{-2} s^{-1} with a median of 8.5×10^{-17}. Six galaxies were observed on two plates or nights, in different fibres, and give the same lines and redshift each time.

One hundred and twenty-two of the 154 have a Legacy Surveys DR10 source within 1.5″, of $r = 21.8$ to 24.5 and median 23.1: 56 fitted as round exponentials, 38 as point sources, 24 as exponentials, three as de Vaucouleurs profiles and one as a Sérsic profile. The 32 without a source lie below the catalogue's depth, so their lines are all the imaging and spectroscopy know of them. None has a DESI DR1 redshift within 1.5″: DESI's targets are chosen from the same imaging, and the sky fibres' galaxies lie below the selection.

3.4 What remains

At 8σ, 10 peaks and 10 dips lie in spectra that hold clean lines of both signs, and 485 peaks and 336 dips are neither fixed to a fibre, at a galaxy's redshift nor in such a spectrum. They are single lines in single fibres: a galaxy whose second line falls on an airglow line, below threshold or beyond the spectrum, a residual of the sky subtraction, or a transmitter would each appear so. Legacy Surveys sources lie within 1.5″ of 181 of the 480 peaks the catalogue covers against 21 at positions

30″ north, and of 10 of the 330 dips against 28; scaled to every line, 162 of the peaks are single lines of galaxies the imaging catalogues, while the dips fall at catalogued sources less often than at random positions. Net of the counterparts, 323 peaks remain against the 336 dips, or 354 with the dips' shortfall at sources restored, so the peaks fall short of the dips by 13 to 31, with an error of 33. Under the one-fibre rule the same counts give 188 peaks against 236 to 245 dips.

The excess of peaks before the counterparts are counted lies at widths of 1.1 to 1.7 pixels, the instrumental profile and a little wider, 210 peaks against 101 dips, where a galaxy's line falls; below 0.9 pixels, where a cosmic ray the pipeline leaves unmasked would fall, 10 peaks lie against 13 dips. The width test alone does not set aside a single-pixel spike, since the second moment of the positive part of the profile carries the noise beside it: in the package's end-to-end test a one-pixel spike at 16σ on synthetic noise passes at a width of 0.82 pixels. The dips' tail is the longer, 47 above 16σ against 27 peaks, many of them where the blue and red cameras' spectra are joined near 6000 Å. Between 6 and 8σ the dips outnumber the peaks, 1899 against 1571, so the residuals do not make the two signs equally at low significance, and the comparison is drawn at 8σ, where they do to within the counts.

3.5 Exposure by exposure

Read in the frames of their exposures, 179 of the 188 galaxy lines at 8σ are persistent, seven unclear and two carried by one exposure or one camera, as lines on the sky must be, while four of the 19 dips at a galaxy's redshift are. Of the 485 peaks that remain, 359 are persistent, 92 unclear, 25 carried by one exposure and one by one camera, and eight lie on fewer than two frames; of the 336 dips, 241 are persistent, 35 unclear, 37 carried by one exposure and 16 by one camera, and seven lie on fewer than two frames. Legacy Surveys sources lie within 1.5″ of 175 of the 358 persistent peaks the catalogue covers against 18 at the control positions, and of 7 of the 236 persistent dips against 19, so net of the counterparts 202 persistent peaks stand against 253 persistent dips, 52 ± 29 short.

The six strongest peaks that remain, at 36 to 52σ, have no Legacy Surveys source. Four are carried by one exposure and one lies on a single frame. The sixth, at 4961.4 Å in fibre 562 of plate 5294, is [O III] 4959 of the planetary nebula Abell 31 (PN G219.1+31.2), whose spectrum fills the fibre from [Ne III] 3869 to [S III] 9531 at zero redshift: its brighter lines reach 3σ in two or more of the plate's other sky fibres and are set aside as shared, the sky fibres off the nebula hold them as dips where the sky model took in its light, and five fainter lines, Hγ, [O III] 4959, [Ar III] 7136 and [S III] 9069 and 9531, remain among the peaks.

3.6 Lines that recur at one position

Clean lines fall on two or more plate-nights at one position at 53 positions among the peaks and 28 among the dips. One line recurs at 37 of the peak positions and at three of the dip positions, and the recurring peaks are 70 of the 485 at 8σ, against three dips. Five of the 37, all between 7204 and 7218 Å, recur at another position as well and are the sky's.

The line found most often lies at RA 213.5584°, Dec +53.6108° in the field of the SDSS reverberation-mapping project [15], where plates 7338, 7339 and 7340, observed on many nights, and three others put a sky fibre in 94 spectra between 2013 and 2020. A line at 4732.6 Å is detected in 81 of them, clean in 54 and among the peaks that remain at 8σ in 41, with no Legacy Surveys source at the fibre. The 94 spectra stacked by inverse variance hold it at 52σ, centred at 4732.7 Å with a flux of 8.5×10^{-17} erg cm^{-2} s^{-1} and a width of 287 km s^{-1} against the 179 km s^{-1} of the instrumental profile, and no other line. Read as [O II] at $z = 0.269$, it puts Hβ, [O III] 5007 and Hα at 6172, 6357 and 8333 Å, where the stack holds −0.02, 0.00 and 0.00 of its flux with errors of 0.01,

while a star-forming galaxy's Hα is as bright as its [O II] or brighter. Read as C IV at $z = 2.054$, it puts He II and C III] where the stack holds -0.01 and 0.01 ± 0.01, and read as Mg II at $z = 0.690$, it puts [O II] and [O III] where the stack holds -0.06 ± 0.03 and 0.05 ± 0.02; an active nucleus would show those lines, and its Mg II is broad. Read as Lyα at $z = 2.893$, it puts N V, C IV, He II and C III] where the stack holds 0.00 to 0.01 ± 0.01, lines that star-forming galaxies at $z \approx 3$ show at a few per cent of Lyα [16], and its intrinsic width, about $220\,\mathrm{km\,s^{-1}}$, is that of a Lyα emitter: a galaxy at $z = 2.893$ below the imaging's depth, resolved where a transmitter's line is not.

The second, [O III] 5007 at 7775.0 Å, recurs at RA 29.4755°, Dec +27.5883° on all seven plate-nights of plates 7691 and 10000 that hold a clean line there, with Hβ beside it on one, which places it in Table 2 at $z = 0.5525$. The other 35 positions repeat a line on two or three plate-nights, and 15 of their 26 lines at 8σ have a Legacy Surveys source within $1.5''$ against one at the control positions.

3.7 Sensitivity

The line flux at the 6σ threshold, read from each clean peak's flux and significance, has a median of $6.0 \times 10^{-17}\,\mathrm{erg\,cm^{-2}\,s^{-1}}$ and a 10–90 per cent range of 3.4×10^{-17} to 2.7×10^{-16}: 6.2×10^{-17} from 3600 to 5000 Å, 5.0×10^{-17} from 5000 to 7000, 4.8×10^{-17} from 7000 to 9000 and 1.5×10^{-16} beyond 9000 Å, where the airglow lines crowd. At 8σ the median is $8.0 \times 10^{-17}\,\mathrm{erg\,cm^{-2}\,s^{-1}}$. An isotropic source giving that flux has a luminosity of 9.5×10^{16} W at 10 pc, 9.5×10^{18} W at 100 pc and 9.5×10^{20} W at 1 kpc, and a transmitter aimed at Earth through a diffraction-limited 10 m aperture at the median wavelength of 7013 Å needs 37 W, 3.7 kW and 370 kW; its beam is 0.14 au wide at 10 pc and 1.4 au at 100 pc, so within 100 pc it must be held on the Earth's position at arrival, and beyond it a beam on the Sun covers the Earth as well.

Lines of the instrumental profile injected into the sky fibres of 40 plates spread through the list, one per fibre at a random unmasked pixel off the airglow lines and 1727 at each strength, are recovered as clean peaks under the two-fibre rule in 56 per cent of cases at the threshold and 70 to 74 per cent at 1.5 to 5 times it, and under the one-fibre rule in 30 and 36 to 38 per cent; the loss is the test against the other fibres, whose residuals reach 3σ within a pixel of a given position in some fibre often enough to read a line as shared. On synthetic plates of white noise the same detector recovers lines at 10σ in 99 and 84 per cent of cases under the two rules, so the loss in the data is the residuals'. At 8σ, one third above the threshold, the two-fibre rule recovers about 0.68 of lines.

The fibres are $2''$ across and cover $0.093\,\mathrm{deg^2}$ in all, so the search bounds the density of transmitters on the sky rather than any chosen star. The excess of peaks over dips at 8σ, below zero with an error of 33, puts the number of transmitter lines below 54 at 95 per cent confidence, the larger of zero and the excess plus 1.645 times its error, and with 0.68 of lines recovered, fewer than about 850 transmitters per square degree give lines above $8 \times 10^{-17}\,\mathrm{erg\,cm^{-2}\,s^{-1}}$ at these wavelengths, 3.5×10^7 over the sky. A transmitter that shines through a plate's exposures persists in its frames as 95 per cent of the galaxies' lines do, which leaves 0.65 of its lines recovered at 8σ, and the persistent peaks, 52 ± 29 short of the persistent dips, put fewer than 47 such lines in the data and fewer than about 790 such transmitters on each square degree.

Table 2: The 154 galaxies and two Galactic fibres: plate, MJD and fibre; the fibre's position; the redshift; the lines found at it; the strongest line's signal-to-noise ratio; and the Legacy Surveys DR10 source within 1.5″, with its profile type and r.

Plate-MJD-fibre	RA	Dec	z	Lines	S/N	Legacy Surveys
8169-57071-296	127.14855	+57.12516	-0.0002	[N II], [O III] 5007, [S II] 6731	7.5	–
6602-56571-766	24.02404	+34.10890	0.0001	[O II], [S II] 6716	8.2	outside
11288-58401-543	351.36997	+6.09305	0.0164	Hα, [O III] 5007	8.7	–
4020-55332-26	226.04718	-2.39425	0.0954	Hα, [O III] 5007	13.5	–
4563-55617-722	154.88368	+38.46850	0.1085	[O III] 4959, [O III] 5007	14.9	–
8175-57039-450	150.61150	+50.64228	0.1416	Hα, [O III] 5007	10.8	round exp., $r = 22.5$
7512-56777-45	134.07495	+48.20114	0.1633	Hα, [O III] 5007	13.6	–
3877-55365-396	225.00402	+28.04808	0.1817	Hβ, [O III] 5007	10.4	round exp., $r = 23.3$
6013-56074-554	213.58866	+24.33099	0.1895	Hα, [O III] 5007	7.5	point, $r = 23.6$
4057-55357-108	245.33337	+20.30591	0.1997	Hβ, [O III] 5007	10.5	exp., $r = 22.5$
9435-58108-594	17.53466	+4.66545	0.1999	Hα, [O III] 5007, [O II]	10.8	round exp., $r = 22.3$
4279-55508-88	357.85425	+2.95924	0.2248	Hα, [O III] 5007	16.3	round exp., $r = 22.2$
7690-57046-103	28.95435	+25.09805	0.2249	Hα, [O III] 5007	13.1	round exp., $r = 22.3$
5378-56011-408	174.56869	+14.12900	0.2277	[O III] 4959, [O III] 5007	19.4	–
7822-57041-354	41.57260	-1.06696	0.2313	Hα, [O II]	7.2	exp., $r = 22.0$
6793-56430-248	242.35395	+54.12879	0.2351	Hα, [O III] 5007	9.5	exp., $r = 22.5$
5404-56013-406	187.52239	+14.68962	0.2357	Hα, [O III] 5007	11.3	round exp., $r = 24.0$
11061-58428-6	26.43771	+11.21563	0.2360	Hα, [O III] 5007	7.7	round exp., $r = 22.5$
5873-56035-868	158.02560	+22.15247	0.2445	Hα, [O III] 5007, [O II]	8.9	round exp., $r = 23.4$
6689-56396-676	177.85168	+54.09125	0.2483	Hα, [O III] 5007	7.8	round exp., $r = 22.3$
11352-58456-248	156.58603	+27.53902	0.2508	Hα, [O III] 5007	12.6	round exp., $r = 23.3$
7311-57038-991	132.07073	+50.44043	0.2680	Hα, [O III] 5007	12.8	–
7512-56777-675	132.07073	+50.44043	0.2680	Hα, [O III] 5007	13.3	–
7699-57574-95	355.68318	+25.99462	0.2812	Hα, [O III] 5007	14.1	exp., $r = 22.6$
10901-58397-179	255.53170	+32.14931	0.2857	Hα, [O III] 5007	9.7	exp., $r = 22.6$
7835-56986-322	33.83880	-3.32220	0.2939	[O III] 4959, [O III] 5007	14.6	round exp., $r = 23.3$
7390-57130-18	163.85952	+47.34356	0.2948	[O III] 5007, [O II]	13.2	round exp., $r = 22.8$
5046-56243-614	343.65339	+13.46747	0.2988	[O III] 5007, [O II]	8.0	round exp., $r = 22.7$
5002-55710-284	261.59575	+30.59913	0.3008	Hα, [O III] 5007	10.1	round exp., $r = 22.3$
9575-57786-786	153.75520	+17.37566	0.3050	Hα, Hβ, [O III] 4959, [O III] 5007	16.6	–
11561-58485-827	2.04539	+11.85705	0.3056	Hα, [O III] 5007	9.0	point, $r = 23.1$
4040-55605-420	207.18925	-0.83451	0.3062	Hα, [O III] 5007	7.3	round exp., $r = 23.5$
5141-55746-113	318.86697	+0.91202	0.3092	Hα, [O III] 5007	9.2	exp., $r = 22.5$
7057-56593-810	41.11542	-7.41403	0.3110	Hα, Hβ, [O III] 4959, [O III] 5007, [O II]	19.0	Sérsic, $r = 22.1$
4740-55651-178	174.48659	+1.11383	0.3113	Hα, Hβ, [S II] 6716	18.6	round exp., $r = 21.8$
7900-57358-798	4.13084	-3.05673	0.3125	Hα, [O III] 5007	12.5	point, $r = 23.5$
10440-58138-745	141.43832	+31.17449	0.3134	[O III] 4959, [O III] 5007	20.4	point, $r = 23.6$
9141-57663-653	321.07987	-0.46193	0.3201	Hβ, [O III] 4959, [O III] 5007, [O II]	18.2	round exp., $r = 22.6$
6743-56385-218	202.05611	+47.97118	0.3268	Hα, [O III] 4959, [O III] 5007	16.4	point, $r = 24.1$
11109-58523-950	124.27555	+22.24246	0.3279	Hα, [O III] 5007	9.0	round exp., $r = 22.2$
4174-55659-442	255.95617	+21.05868	0.3289	Hα, [O III] 5007	7.8	round exp., $r = 23.5$
5198-55823-408	252.65262	+35.90116	0.3395	[O III] 5007, [O II]	13.2	round exp., $r = 23.9$
8370-57520-708	175.74678	+44.52171	0.3412	Hα, [O III] 5007	12.1	round exp., $r = 23.4$

Plate-MJD-fibre	RA	Dec	z	Lines	S/N	Legacy Surveys
8370-57732-718	175.74678	+44.52171	0.3412	Hα, [O III] 5007	10.1	round exp., $r = 23.4$
10738-58251-439	240.90571	+35.18927	0.3416	Hα, [O II]	12.0	round exp., $r = 22.7$
7095-56625-482	160.11325	+58.20823	0.3498	[O III] 4959, [O III] 5007, [O II]	23.4	round exp., $r = 22.6$
6587-56537-574	349.59914	+27.54950	0.3516	[O III] 5007, [O II]	9.0	exp., $r = 22.1$
10917-58252-746	241.41881	+32.47987	0.3524	Hα, [O III] 5007	8.5	round exp., $r = 23.5$
6633-56369-490	183.16471	+41.81318	0.3568	[O III] 4959, [O III] 5007	15.3	round exp., $r = 23.6$
4543-55888-240	11.54328	+6.28194	0.3584	[O III] 5007, [O II]	7.6	point, $r = 23.2$
9382-57746-452	34.84357	+0.03436	0.3624	[O III] 5007, [O II]	14.8	de Vauc., $r = 22.2$
6586-56485-366	336.95291	+28.14091	0.3655	[O III] 5007, [O II]	9.8	round exp., $r = 21.9$
6289-56559-308	348.17625	+27.40934	0.3717	[N II], [O II]	8.4	point, $r = 22.7$
10748-58224-776	223.64392	+38.18774	0.3775	[O III] 5007, [O II]	7.7	round exp., $r = 23.4$
4643-55946-153	138.35170	+37.83216	0.3833	Hα, [O III] 5007	9.5	round exp., $r = 22.8$
4016-55632-414	225.44444	-0.61182	0.3888	Hα, [O III] 5007	8.6	–
9442-58076-984	9.95057	+2.65555	0.3919	Hα, [O III] 4959, [O III] 5007	14.7	–
7701-57356-625	351.51853	+27.44095	0.3932	Hα, [O III] 5007	10.2	–
11059-58515-987	23.81695	+17.02814	0.3960	Hα, [O II]	11.2	round exp., $r = 22.8$
9162-58040-242	322.00060	+0.38206	0.3972	Hα, [O III] 5007	9.5	point, $r = 23.1$
7631-57279-635	18.95434	+22.18753	0.3977	Hα, [O III] 5007	10.2	–
8526-57898-317	244.45013	+41.51516	0.3982	Hα, Hβ, [O III] 4959, [O III] 5007	14.5	point, $r = 24.0$
6044-56090-107	244.45014	+41.51515	0.3983	Hα, [O III] 4959, [O III] 5007	17.6	point, $r = 24.0$
8792-57364-446	23.51387	-1.05226	0.4020	Hα, [O III] 5007	7.0	round exp., $r = 23.6$
4656-55828-488	28.11478	+11.68275	0.4028	Hα, [O III] 5007	8.4	point, $r = 23.7$
5694-56213-351	23.77663	+25.07137	0.4034	Hα, [O II]	7.7	–
3878-55361-470	221.72976	+28.42111	0.4085	Hα, [O III] 5007	8.7	round exp., $r = 23.4$
8211-57423-986	205.78650	+50.99986	0.4098	Hα, [O III] 5007	12.0	–
6423-56313-38	178.98923	+22.51120	0.4127	Hα, [O III] 5007	10.7	round exp., $r = 23.1$
4038-55363-444	210.29613	-3.23692	0.4129	Hα, [O III] 5007	14.7	exp., $r = 22.6$
5777-56280-948	140.43369	+26.23998	0.4152	Hα, [O III] 5007	7.0	round exp., $r = 22.7$
8750-58401-690	10.33064	+6.25421	0.4302	[O III] 5007, [O II]	15.2	point, $r = 23.3$
9554-57807-188	140.49693	+18.22247	0.4315	Hα, [O III] 5007	8.5	point, $r = 23.6$
5307-55998-786	140.49693	+18.22247	0.4315	Hα, [O III] 5007	10.2	point, $r = 23.6$
6402-56334-335	178.75863	+27.52210	0.4372	[O III] 4959, [O III] 5007	12.4	point, $r = 23.9$
6588-56536-822	344.67859	+25.91339	0.4374	[O III] 5007, [O II]	9.4	round exp., $r = 23.2$
8732-57418-829	35.93424	+5.21363	0.4385	[O III] 4959, [O III] 5007	20.8	point, $r = 23.6$
7114-56748-520	169.81510	+65.92926	0.4398	[O III] 5007, [O II]	11.6	round exp., $r = 23.0$
11554-58456-170	353.86311	+9.93507	0.4408	[O III] 5007, [O II]	9.9	point, $r = 23.7$
6165-56208-788	353.86311	+9.93508	0.4408	[O III] 4959, [O III] 5007	16.5	point, $r = 23.7$
4641-55947-314	138.90323	+40.53916	0.4416	[O III] 5007, [O II]	12.2	exp., $r = 22.4$
7646-57570-208	332.07143	+27.39413	0.4485	[O III] 4959, [O III] 5007, [O II]	15.1	point, $r = 23.8$
6487-56362-880	198.02095	+31.92434	0.4487	[O III] 4959, [O III] 5007, [O II]	12.7	point, $r = 23.3$
10914-58257-755	248.98563	+33.41463	0.4504	[O III] 5007, [O II]	12.7	round exp., $r = 22.4$
11380-58462-704	150.33497	+31.84991	0.4564	[O III] 5007, [O II]	7.5	round exp., $r = 23.2$
7391-56781-675	160.40951	+46.60769	0.4672	Hα, [O III] 5007	13.1	round exp., $r = 22.9$
6516-56571-602	355.05918	+27.08038	0.4683	Hβ, [O III] 5007	13.4	round exp., $r = 23.4$
8497-58161-40	221.93867	+42.20213	0.4761	Hβ, [O III] 5007	31.3	–
6203-56266-288	12.00438	+14.31752	0.4827	[O III] 5007, [O II]	6.9	round exp., $r = 22.0$

Plate-MJD-fibre	RA	Dec	z	Lines	S/N	Legacy Surveys
6699-56411-882	169.24331	+52.16521	0.4856	Hα, [O III] 5007, [O II]	8.6	point, $r = 23.0$
9610-58137-172	133.00640	+27.21546	0.4895	[O III] 5007, [O II]	7.0	point, $r = 22.8$
6634-56367-520	179.94605	+41.95528	0.4895	Hα, [O III] 5007	8.4	point, $r = 23.0$
6484-56358-594	193.85918	+31.69102	0.4952	[O III] 4959, [O III] 5007	14.0	–
11071-58429-332	24.19181	+6.40044	0.5064	[O III] 5007, [O II]	9.2	round exp., $r = 23.1$
8279-57063-174	127.02745	+43.19890	0.5109	[O III] 5007, [O II]	7.3	round exp., $r = 23.3$
7107-56740-796	174.80896	+62.15769	0.5146	[O III] 5007, [O II]	9.1	point, $r = 23.1$
7606-56977-311	350.26431	+21.01340	0.5177	Hβ, [O II]	13.6	exp., $r = 22.1$
7860-57006-380	22.15587	+1.18941	0.5261	Hβ, [O III] 5007	6.8	exp., $r = 21.9$
8204-58146-416	198.21603	+55.42233	0.5277	[O III] 5007, [O II]	7.7	exp., $r = 22.0$
4994-55739-370	258.61404	+33.83769	0.5293	[O III] 4959, [O III] 5007	18.7	–
6422-56328-90	176.49067	+21.57333	0.5318	Hβ, [O III] 4959, [O III] 5007, [O II]	32.5	point, $r = 23.4$
3847-55298-412	185.87047	-0.69507	0.5432	Hβ, [O III] 4959, [O III] 5007	21.5	–
3847-55588-418	185.87047	-0.69507	0.5432	Hβ, [O III] 4959, [O III] 5007	20.6	–
3868-55360-202	219.81346	+30.49225	0.5504	Hβ, [O II]	11.9	exp., $r = 22.3$
7691-58487-687	29.47546	+27.58831	0.5525	Hβ, [O III] 5007	13.1	–
9144-57666-648	328.28503	+0.02365	0.5563	Hβ, [O II]	14.4	round exp., $r = 22.0$
5313-55973-558	142.25359	+13.82727	0.5702	[O III] 5007, [O II]	8.7	exp., $r = 22.7$
4747-55652-835	180.80191	+4.98357	0.5885	[O III] 5007, [O II]	12.0	exp., $r = 22.3$
10752-58488-557	217.96798	+39.67842	0.5980	Hβ, [O III] 5007	6.9	–
3947-55332-324	235.38928	+23.87738	0.6051	[O III] 5007, [O II]	8.3	exp., $r = 22.3$
11337-58431-190	330.43338	+5.56656	0.6088	[O III] 5007, [O II]	6.9	exp., $r = 22.7$
3940-55327-364	239.43312	+23.99639	0.6203	Hβ, [O III] 4959, [O III] 5007	28.3	exp., $r = 22.6$
4533-55855-414	2.28310	+7.12388	0.6208	[O III] 4959, [O III] 5007, [O II]	20.1	round exp., $r = 22.8$
8755-58138-198	25.24367	+5.87333	0.6300	[O III] 5007, [O II]	8.2	round exp., $r = 22.5$
6155-56245-438	347.48707	+10.35645	0.6312	[O III] 4959, [O III] 5007	13.7	–
5191-56065-388	245.24949	+34.60598	0.6395	[O III] 4959, [O III] 5007	11.8	–
9582-57806-566	143.08944	+17.21156	0.6396	[O III] 5007, [O II]	9.1	de Vauc., $r = 22.4$
3781-55243-146	141.78061	-2.11227	0.6453	[O III] 5007, [O II]	11.7	point, $r = 23.1$
9162-58040-928	322.94891	+1.91979	0.6476	[O III] 4959, [O III] 5007	16.9	point, $r = 23.9$
4779-55656-64	221.47363	+3.87355	0.6515	[O III] 5007, [O II]	8.0	exp., $r = 22.5$
4753-55674-152	188.64795	+3.56641	0.6634	[O III] 5007, [O II]	12.4	round exp., $r = 22.9$
10429-58145-627	143.02481	+32.28740	0.6823	[O III] 5007, [O II]	13.1	point, $r = 23.0$
6497-56329-300	213.29485	+29.07750	0.6937	Hβ, [O III] 5007, [O II]	27.6	exp., $r = 22.5$
4858-55686-359	221.21223	+6.63193	0.7023	[O III] 5007, [O II]	8.5	exp., $r = 23.1$
8388-57488-346	198.47179	+41.25348	0.7037	[O III] 5007, [O II]	7.3	round exp., $r = 23.0$
7747-58398-852	3.14290	+35.45403	0.7039	[O III] 5007, [O II]	7.7	outside
4221-55443-510	7.27573	+0.01951	0.7185	[O III] 5007, [O II]	10.5	exp., $r = 22.9$
5739-55976-150	149.61539	+48.66389	0.7191	[O III] 4959, [O III] 5007	14.0	–
7183-56658-450	25.90411	-10.25796	0.7360	Hβ, [O III] 4959, [O III] 5007	21.7	exp., $r = 22.7$
8234-57450-582	177.12140	+58.45203	0.7437	Hβ, [O III] 5007	8.2	round exp., $r = 22.9$
5863-56042-816	197.31227	+18.68198	0.7451	Hβ, [O III] 5007	8.7	round exp., $r = 22.5$
5130-55835-42	26.45194	+14.74649	0.7455	Hβ, [O III] 5007	19.9	point, $r = 23.6$
4348-55559-700	27.49099	-2.31464	0.7476	[O III] 5007, [O II]	30.7	round exp., $r = 23.3$
8763-57309-452	348.32325	-0.06942	0.7518	Hβ, [O II]	6.7	–
5733-56575-654	138.00488	+48.77073	0.7541	Hβ, [O II]	8.0	–

Plate-MJD-fibre	RA	Dec	z	Lines	S/N	Legacy Surveys
6881-56540-554	2.46761	+23.78729	0.7575	[O III] 4959, [O III] 5007	16.6	point, $r = 23.4$
6825-56717-460	202.51184	+66.02136	0.7933	Hβ, [O II]	9.1	round exp., $r = 23.2$
5979-56329-226	185.96476	+23.12713	0.7962	Hβ, [O II]	8.9	point, $r = 23.6$
5800-56279-368	149.00528	+29.49835	0.8026	[O III] 5007, [O II]	6.2	round exp., $r = 23.6$
7854-56989-316	5.58985	+1.51164	0.8032	[O III] 5007, [O II]	7.6	de Vauc., $r = 23.1$
4376-55863-124	330.24601	-3.46081	0.8089	[O III] 4959, [O III] 5007	25.1	–
7142-56567-606	338.98836	+33.05879	0.8099	[O III] 4959, [O III] 5007	19.7	point, $r = 24.0$
4207-55475-812	345.33907	+0.36194	0.8357	[O III] 5007, [O II]	11.0	point, $r = 24.5$
5413-55976-488	191.68889	+7.49739	0.8365	[O III] 5007, [O II]	7.6	round exp., $r = 23.2$
10757-58223-520	215.56647	+35.41421	0.8369	[O III] 5007, [O II]	6.8	point, $r = 23.8$
4478-55600-376	126.61986	+21.27518	0.8550	[O III] 5007, [O II]	8.1	point, $r = 23.7$
5410-56016-644	191.05282	+14.52098	0.8552	[O III] 5007, [O II]	6.8	point, $r = 24.0$
5447-55958-124	210.25983	+7.91491	0.8552	[O III] 5007, [O II]	6.3	round exp., $r = 23.4$
4447-55542-449	122.63066	+29.97605	0.8795	[O III] 5007, [O II]	8.5	point, $r = 24.2$
5703-56190-288	14.54918	+19.37136	0.9006	Hβ, [O II]	15.1	point, $r = 22.9$
3537-55116-602	318.00184	+0.45669	0.9352	Hβ, [O III] 5007	7.8	–
6529-56535-447	9.74883	+30.63794	0.9394	Hβ, [O III] 5007	10.6	–
7681-57042-706	18.71776	+24.94611	0.9813	Hβ, [O II]	18.2	exp., $r = 22.3$
6625-56386-450	199.19516	+45.56782	1.0251	[O III] 5007, [O II]	13.1	round exp., $r = 23.2$
6645-56398-490	174.26890	+45.42930	1.0364	[O III] 4959, [O III] 5007	6.8	–
3647-56568-535	35.90259	+0.31456	1.0690	[O III] 4959, [O III] 5007	9.5	–

4 Related work

Spectroscopic laser searches have examined stars: 577 with Keck [1], 5600 with thresholds of megawatts [2] and the Automated Planet Finder's targets [3]. Sky fibres serve the surveys as a measure of the sky, and their residuals are studied to improve its subtraction [14]; the sky spectra themselves have been mined for the diffuse Galactic light, by correlating 92,000 SDSS sky spectra with the far-infrared dust emission [4], 250,000 BOSS sky spectra [5] and 10.8 million DESI sky fibres [6], whose line lists of the airglow and the nebular sky are the reference for what a single sky fibre holds. We find no earlier search of sky fibres for unresolved lines in single fibres, nor a catalogue of the galaxies they record.

5 Conclusion

No unresolved emission line in the 381,943 sky-fibre spectra of the BOSS and eBOSS plates stands above the residuals of the sky subtraction once every test is applied to peaks and dips alike and the single lines of catalogued galaxies are counted: at 8σ, 323 peaks remain against 336 dips, and of the lines that hold through the exposures, 202 peaks against 253 dips. Galaxies and nebulae are what the blank sky holds: 154 galaxies found by two or more lines at one redshift where residuals meet the same test four times, a fifth of them below the Legacy Surveys' depth, two Galactic fibres, the planetary nebula Abell 31, and a Lyα emitter at $z = 2.893$ that one sky position shows on 54 plate-nights. The search bounds a transmitter in a sky fibre at a line flux of $8 \times 10^{-17}\,\mathrm{erg\,cm^{-2}\,s^{-1}}$, kilowatts through a 10 m aperture at 100 pc, and such transmitters to fewer than about 850 per square degree. The same search runs unchanged on the sky fibres of DESI, whose 5000 fibres place

several hundred on blank sky in every tile, and on the integral-field cubes of MaNGA and SAMI, which cover blank sky around every target; the data behind every number here, each peak and dip with its tests and counterparts, are published with the paper.

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